GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

www.dmirs.wa.gov.au/geochron

Department of Mines, Industry Regulation and Safety

231142.1.pdf *Geochronology Record 1914*

231142.1: hornblende–biotite metagranodiorite, Mount Dwyer

(Big Bell Suite, Austin Downs Supersuite; Youanmi Terrane, Yilgarn Craton)

SS Romano, Y Lu, IOH Fielding, MTD Wingate and TJ Ivanic

Location and sampling

YOUANMI SH50-04, EVERETT CREEK 2841 Zone 50, MGA 746202E 6881296N
Warox Site TJISST180046 Sampled on 29 May 2018

This sample was collected from the 327.2 – 327.7 m depth interval of diamond drillcore TDD036, drilled in 2008 by Troy Resources NL at their Sandstone gold project. The drillhole is located about 4.7 km east of Marley Well, 1.7 km south-southeast of Mount Dwyer, and 1.0 km north of the Menzies – Sandstone Road.

Geological context

The unit sampled is a metagranodiorite of the Big Bell Suite (Zibra 2021). The Big Bell Suite is part of the Austin Downs Supersuite and is exposed over a wide area in the northern Youanmi Terrane in the western Yilgarn Craton. It consists mainly of K-feldspar-porphyritic and equigranular monzogranite and were emplaced between c. 2732 and 2686 Ma. The geochemistry of the Big Bell Suite shows a dominant tonalite–trondhjemite–granodiorite (TTG) affinity. The sampled granodiorite has intruded the boundary between the Meekatharra Formation that has been deposited between 2800-2760 Ma and unassigned greenstones of the Sandstone greenstone belt. A metatonalite 13.5 km to the west yielded a maximum crystallization age of 2740 ± 4 Ma (GSWA 229355, Wingate et al. 2020). Two foliated biotite granodiorites sampled about 13.6 and 12.3 km to the north yielded igneous crystallization age of 2686 ± 5 Ma (GSWA 165365, Nelson, 2004) and 2691 ± 3 Ma (GSWA 169070, Nelson, 2002), respectively.

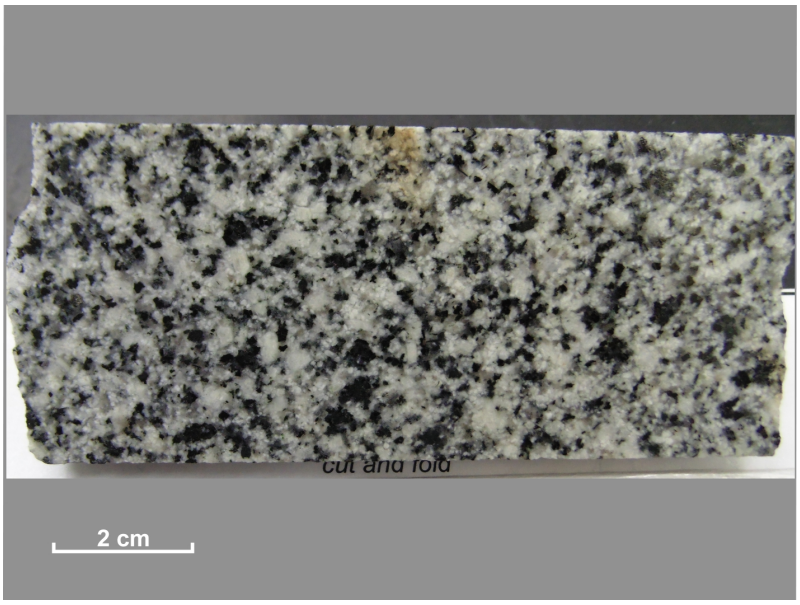

Figure 1. Drillcore image for sample 231142: hornblende–biotite metagranodiorite, Mount Dwyer

Petrographic description

The sample is a medium-grained, equigranular hornblende–biotite metagranodiorite, consisting of about 40–45% plagioclase, 30–35% quartz, 10% K-feldspar, 10% biotite, 3–4% hornblende, 1% magnetite and accessory apatite and zircon. Plagioclase has mostly been saussuritized, whereas biotite has been partially replaced by chlorite and hornblende by actinolite. Plagioclase is tabular, subhedral to anhedral, and up to 2.5 mm in size. Rare relict plagioclase grains show andesine composition (around An_{38}). Quartz is anhedral, up to 2 mm in size with interdigitating grain boundaries and exhibits undulose extinction. K-feldspar is perthitic, occurring as anhedral grains up to 2 mm in size. Brown biotite forms

stubby subhedral grains up to 1.5 mm in size. Subhedral and euhedral hornblende grains up to 1.7 mm in size are pale to olive green. Magnetite occurs as equant, subhedral and rarely euhedral grains that range from <0.1 to 0.3 mm. Apatite forms prismatic euhedral grains to 0.15 mm in length. The granodiorite lacks a tectonic foliation but has undergone solid-state deformation at relatively low grade.

Zircon morphology

Zircons from this sample are colourless to dark brown, and subhedral to euhedral. The crystals are up to 230 µm long, and mainly elongate, with aspect ratios up to 7:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and a few crystals appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

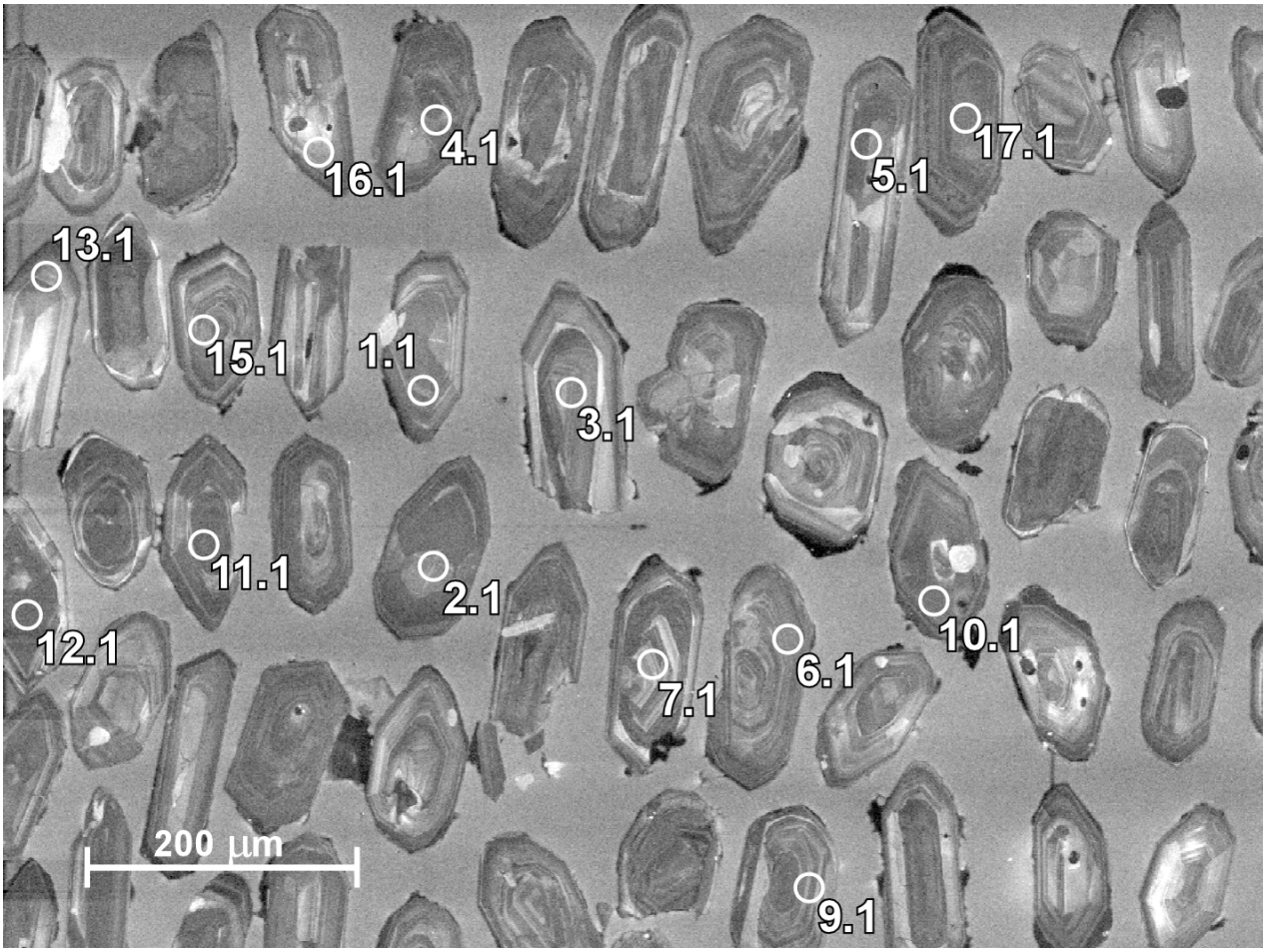

Figure 2. Cathodoluminescence image of representative zircons from sample 231142: hornblende–biotite metagranodiorite, Mount Dwyer. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 21–22 February 2019, using SHRIMP-B. Eleven analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.77% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-three analyses were obtained from 23 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are slightly discordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable and are considered not to be geologically significant. The remaining 21 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 21 analyses of 21 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2701 ± 3 Ma (MSWD = 0.90).

The date of 2701 ± 3 Ma for the 21 analyses in Group I is interpreted as the igneous crystallization age of the granodiorite.

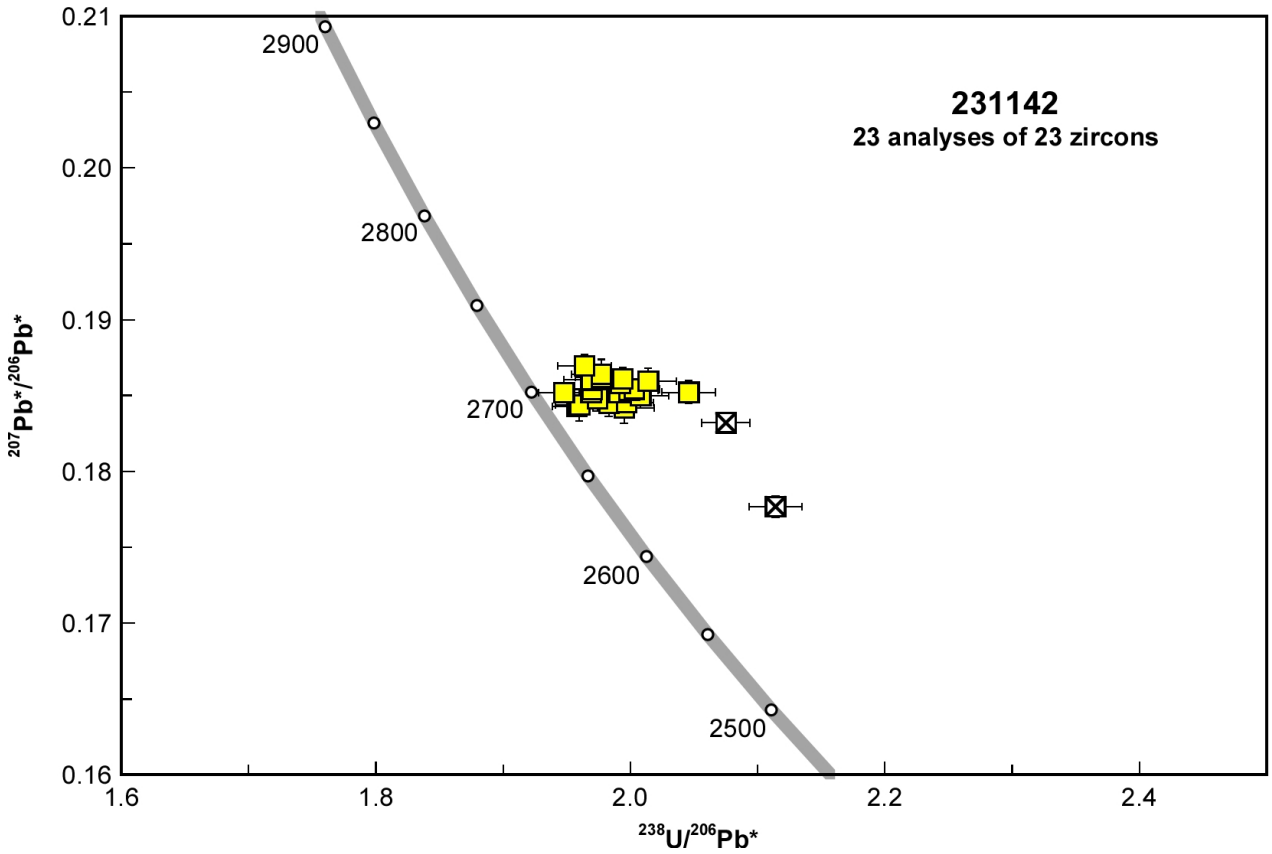

Figure 3. U–Pb analytical data for zircons from sample 231142: hornblende–biotite metagranodiorite, Mount Dwyer. Yellow squares indicate Group I (magmatic zircons); crossed squares indicate Group D (discordance >5%)

Links

Table 1. Ion microprobe analytical results for zircons from sample 231142: hornblende–biotite metagranodiorite, Mount Dwyer: 231142.1.Table1.xls
SHRIMP spot images: 231142_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

References

Nelson, DR 2002, 169070: foliated biotite granodiorite, Coomb Bore; Geochronology Record 108: Geological Survey of Western Australia, 4p. View Reference online

Nelson, DR 2004, 165365: foliated biotite granodiorite, Coomb Bore; Geochronology Record 218: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Wingate, MTD, Lu, Y and Ivanic, TJ 2020, 229355: metatonalite, Lightning Well; Geochronology Record 1684: Geological Survey of Western Australia, 5p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Zibra, I 2021, Big Bell Suite (A-SDB-mg); Geological Survey of Western Australia; WA Geology Online, Explanatory Notes extract, <www.dmp.wa.gov.au/ens>.

Recommended reference for this publication

Romano, SS, Lu, Y, Fielding, IOH, Wingate, MTD and Ivanic, TJ 2023, 231142.1: hornblende–biotite metagranodiorite, Mount Dwyer; Geochronology Record 1914: Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract viewed 11 October 2023. <www.dmirs.wa.gov.au/geochron>

Data obtained: 22 February 2019
Date released: 31 October 2023

This page was last modified on 11 October 2023.

Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

Further details of geoscience products are available from:

Information Centre

Department of Mines, Industry Regulation and Safety

100 Plain Street

EAST PERTH, WA 6004

Telephone: +61 8 9222 3459 | Facsimile: +61 8 9222 3444

www.dmirs.wa.gov.au /GSWApublications

Acknowledgment of Country

The Government of Western Australia acknowledges Aboriginal people as the traditional custodians throughout Western Australia and their enduring connection to the land, waterways and community. We pay our respects to Elders past and present.

Disclaimer

This product uses information from various sources. The Department of Mines, Industry Regulation and Safety (DMIRS) and the State cannot guarantee the accuracy, currency or completeness of the information. Neither the department nor the State of Western Australia nor any employee or agent of the department shall be responsible or liable for any loss, damage or injury arising from the use of or reliance on any information, data or advice (including incomplete, out of date, incorrect, inaccurate or misleading information, data or advice) expressed or implied in, or coming from, this publication or incorporated into it by reference, by any person whosoever.

© State of Western Australia (Department of Mines, Industry Regulation and Safety) 2023
With the exception of the Western Australian Coat of Arms and other logos, and where otherwise noted, these data are provided under a Creative Commons Attribution 4.0 International License. (https://creativecommons.org/licenses/by/4.0/legalcode)